Faculté des Sciences et des Technologies

Laboratoire N° 8 _ Réseaux II

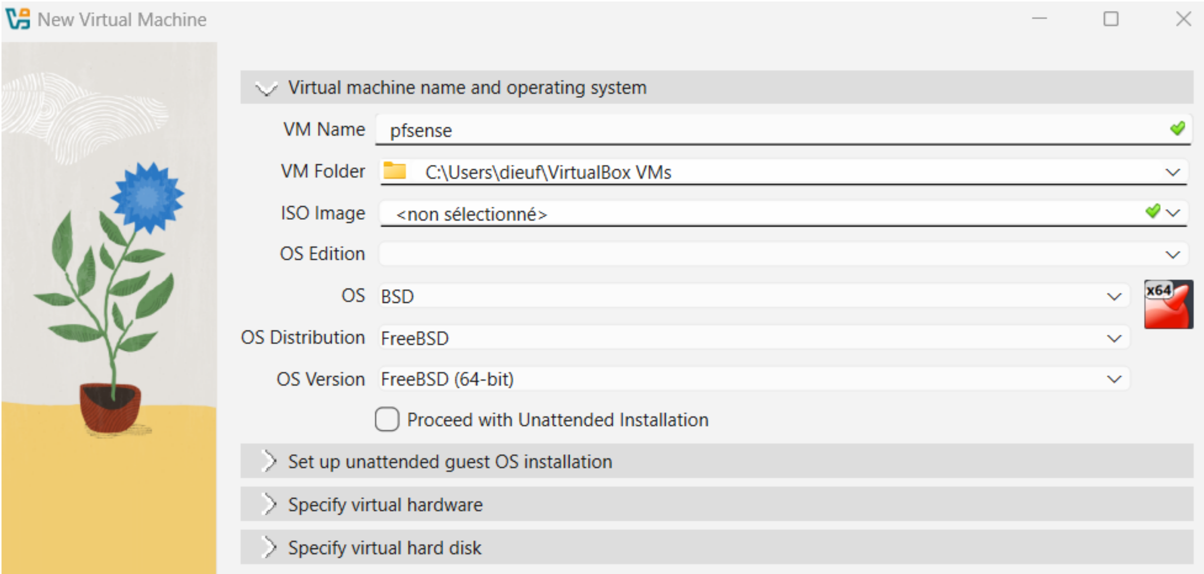
Niveau : Licence III

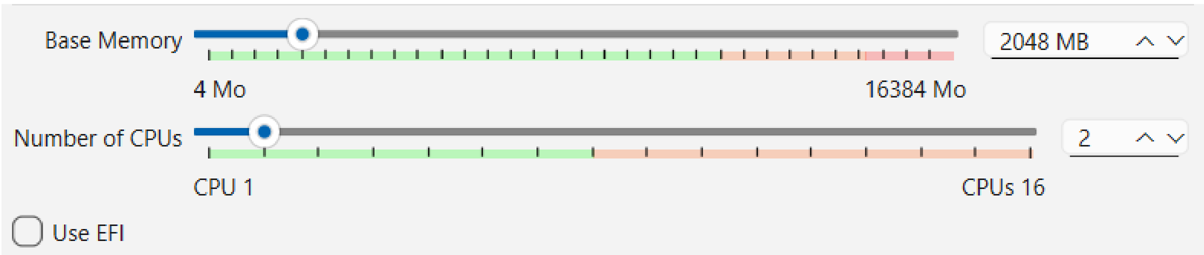
Etudiant : Donsam Jean Gabard NOEL

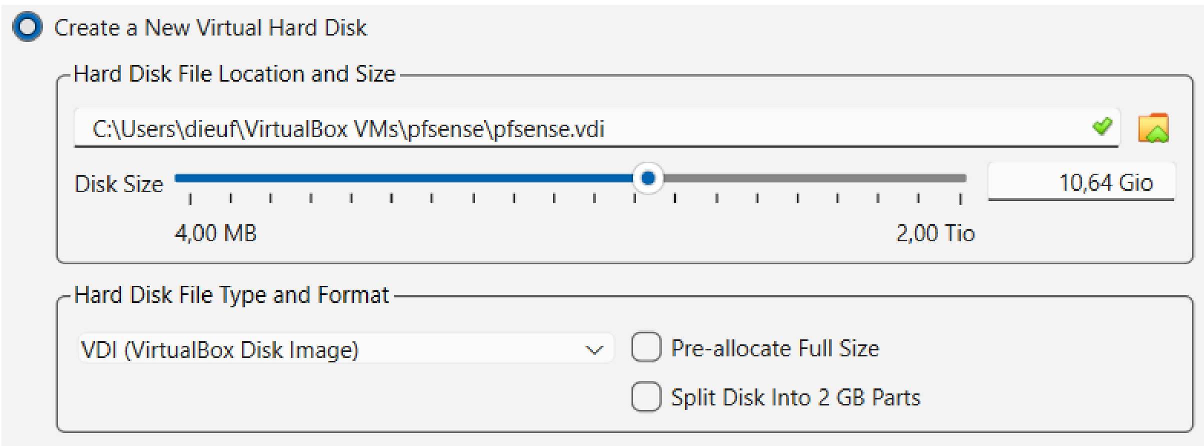
Professeur : Ismaël SAINT ARMOUR

Date : 19 Mai 2026

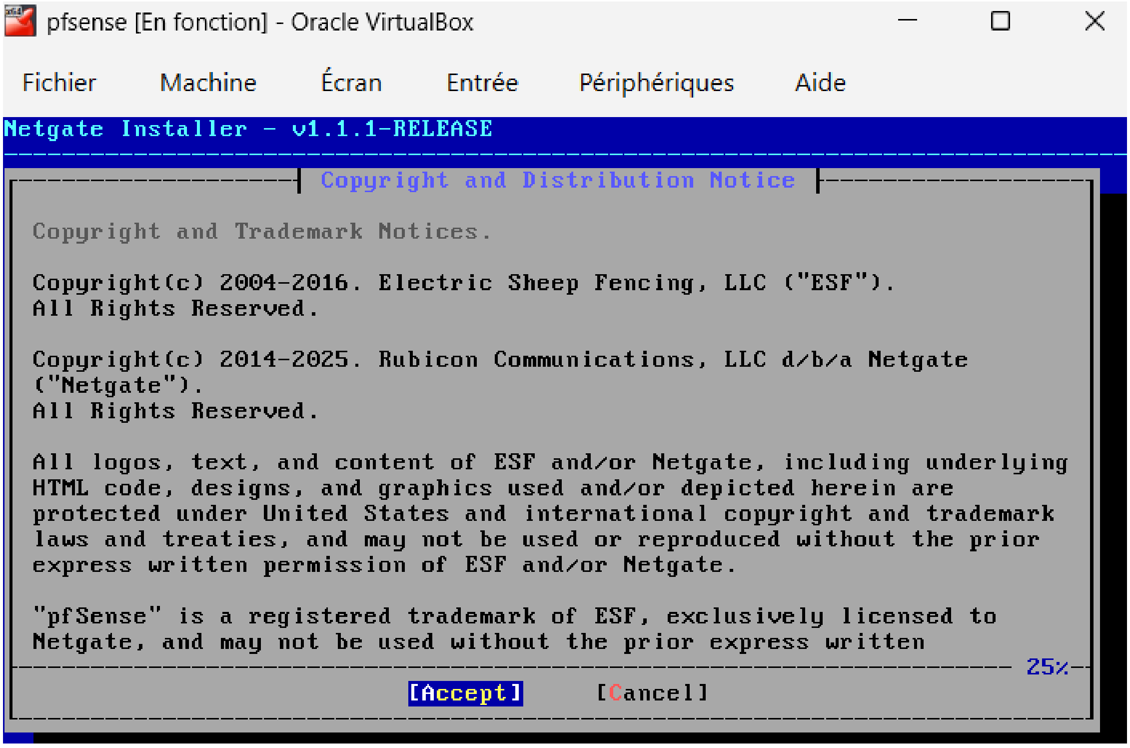
ÉTAPE 1 : Création de la VM (pfsense)

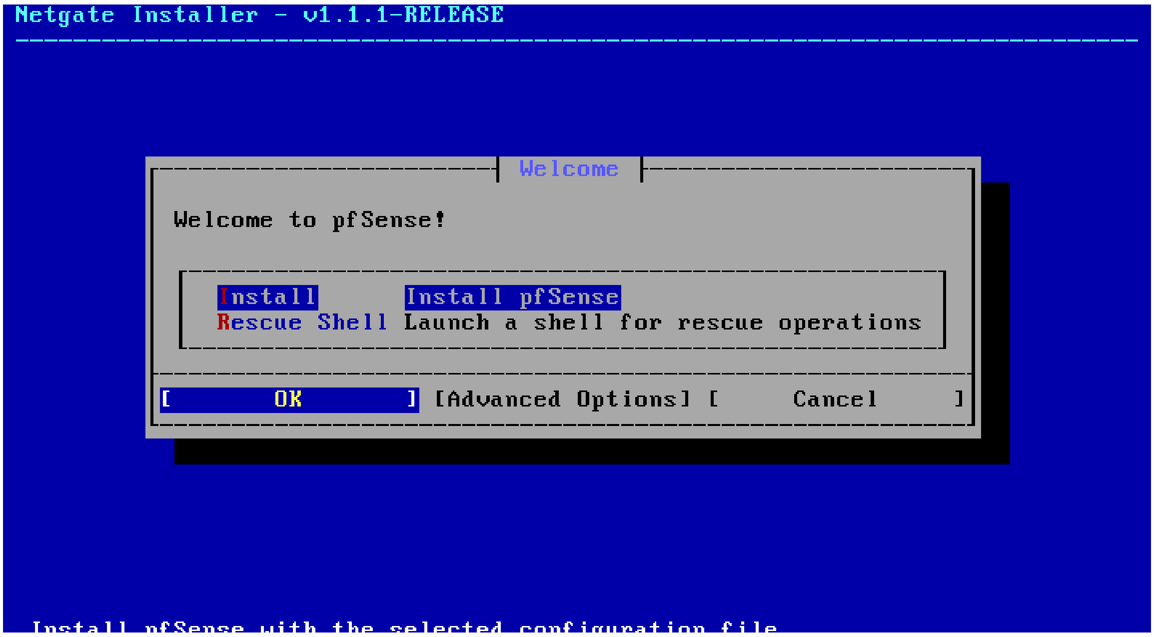


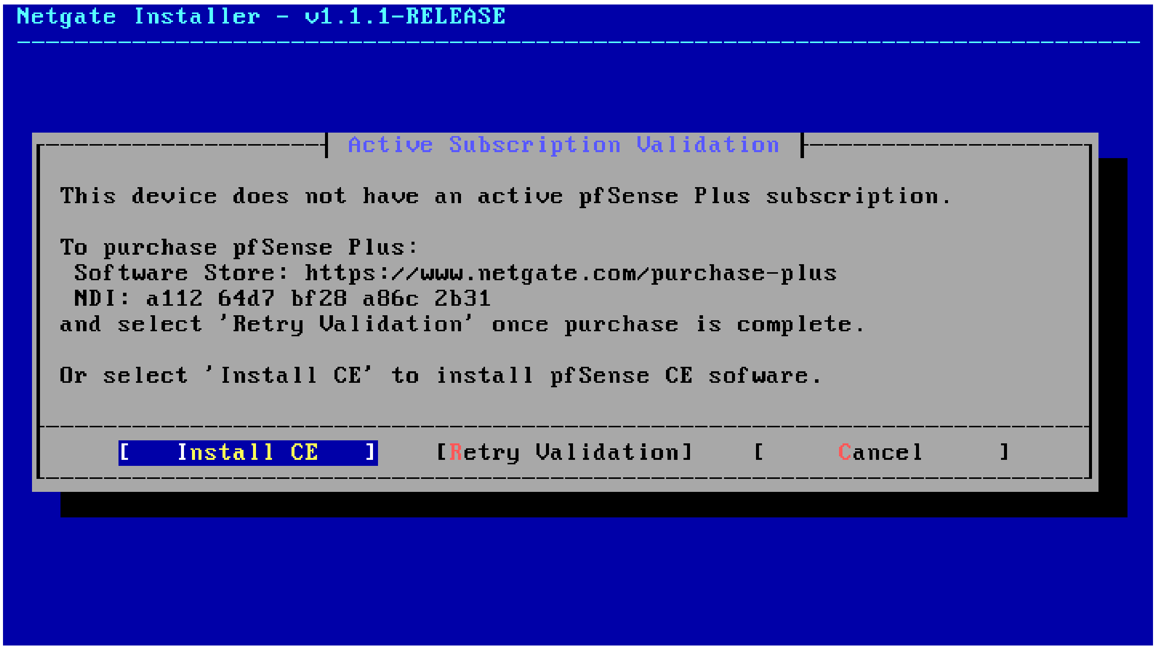


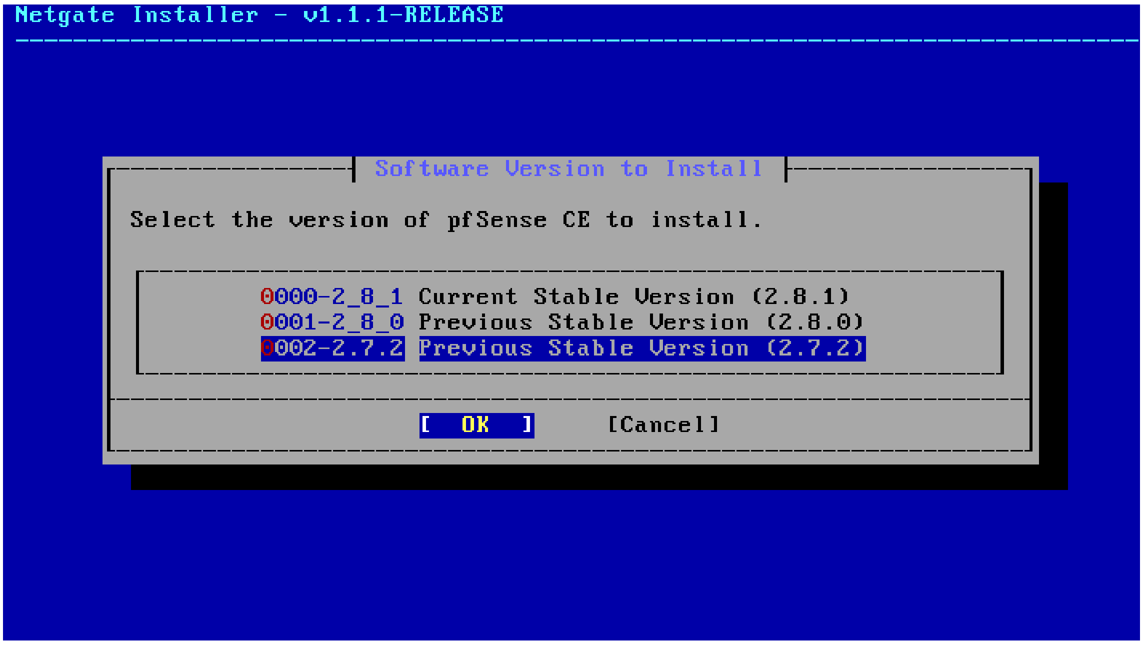


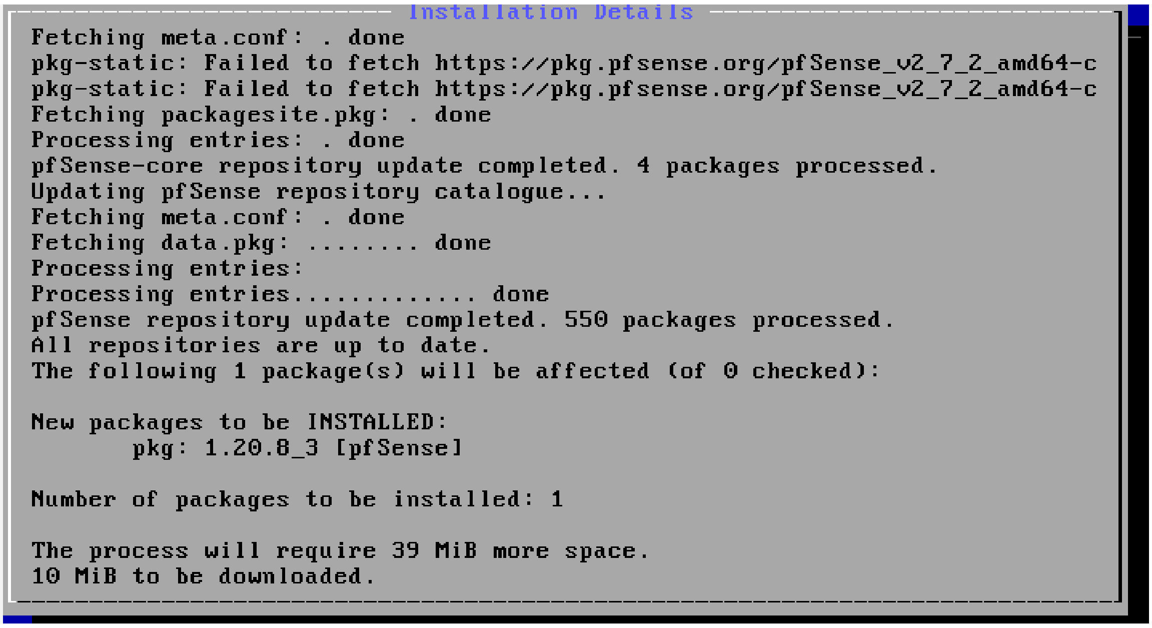
ÉTAPE 2 : Installation de pfSense

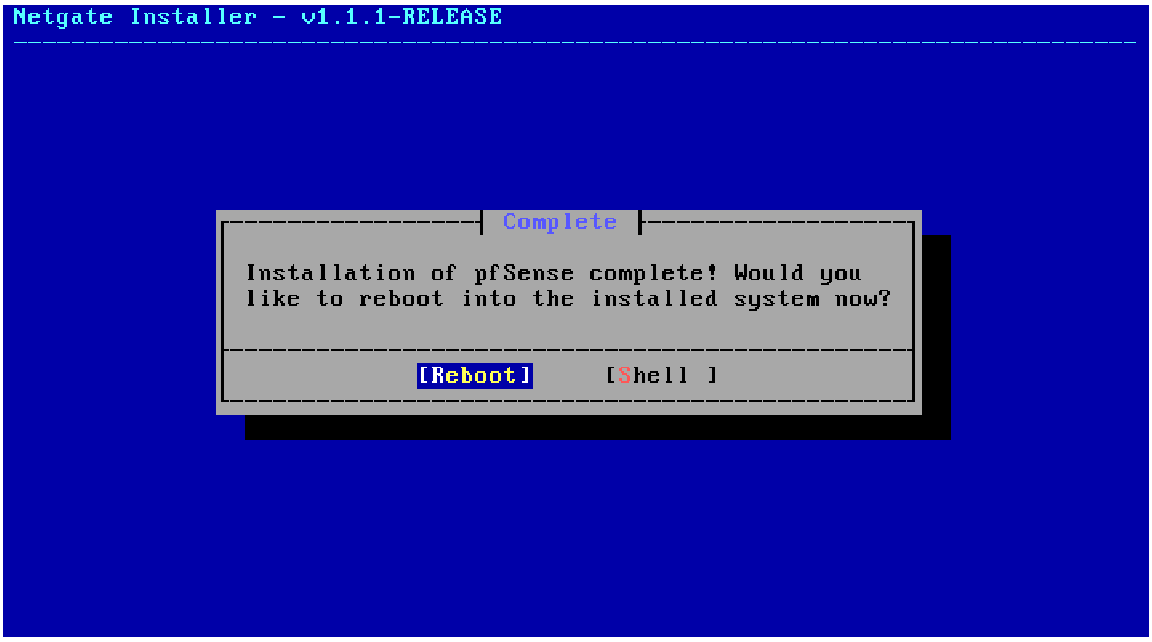





Netgate Installer - v1.1.1-RELEASE
Active Subscription Validation
This device does not have an active pfSense Plus subscription.
To purchase pfSense Plus:
Software Store: https://www.netgate.com/purchase-plus
NDI: a112 64d7 bf28 a86c 2b31
and select 'Retry Validation' once purchase is complete.
Or select 'Install CE' to install pfSense CE sofware.
[Install CE]
[Retry Validation]
[Cancel]


Netgate Installer - v1.1.1-RELEASE
Software Version to Install
Select the version of pfSense CE to install.
0000-2_8_1 Current Stable Version (2.8.1)
0001-2_8_0 Previous Stable Version (2.8.0)
0002-2.7.2 Previous Stable Version (2.7.2)
[OK]
[Cancel]


Installation Details
Fetching meta.conf: . done
pkg-static: Failed to fetch https://pkg.pfsense.org/pfSense_v2_7_2_amd64-c
pkg-static: Failed to fetch https://pkg.pfsense.org/pfSense_v2_7_2_amd64-c
Fetching packagesite.pkg: . done
Processing entries: . done
pfSense-core repository update completed. 4 packages processed.
Updating pfSense repository catalogue...
Fetching meta.conf: . done
Fetching data.pkg: .... done
Processing entries:
Processing entries............. done
pfSense repository update completed. 550 packages processed.
All repositories are up to date.
The following 1 package(s) will be affected (of 0 checked):
New packages to be INSTALLED:
pkg: 1.20.8_3 [pfSense]
Number of packages to be installed: 1
The process will require 39 MiB more space.
10 MiB to be downloaded.


Netgate Installer - v1.1.1-RELEASE
Complete
Installation of pfSense complete! Would you like to reboot into the installed system now?
[Reboot]
[Shell]

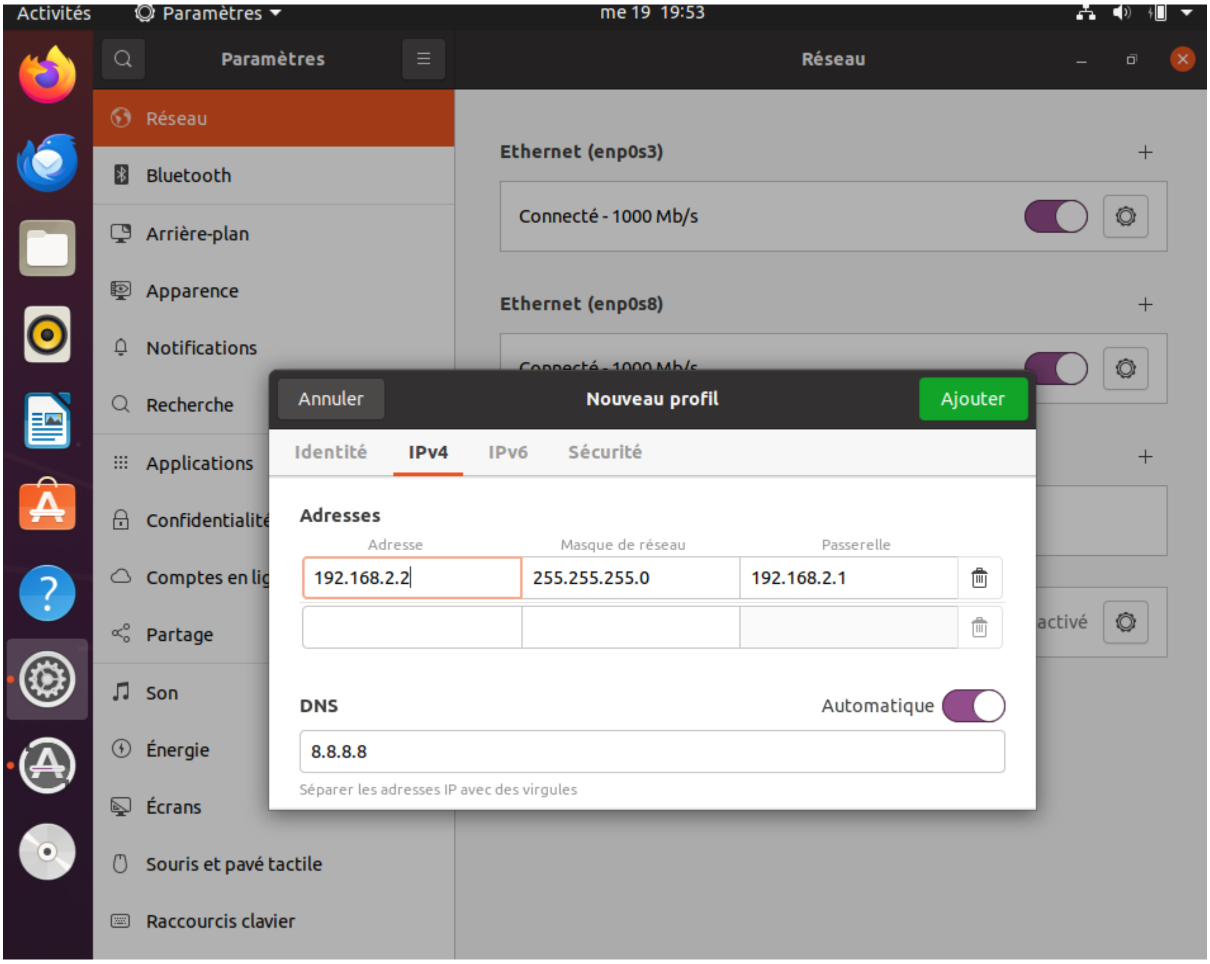

pfSense 2.7.2-RELEASE amd64 20240304-1953
Bootup complete
FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: a11264d7bf28a86c2b31
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:feed:afc/64
LAN (lan) -> em1 -> v4: 192.168.1.1/24
0) Logout (SSH only)
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset webConfigurator password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart webConfigurator
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM
Enter an option:

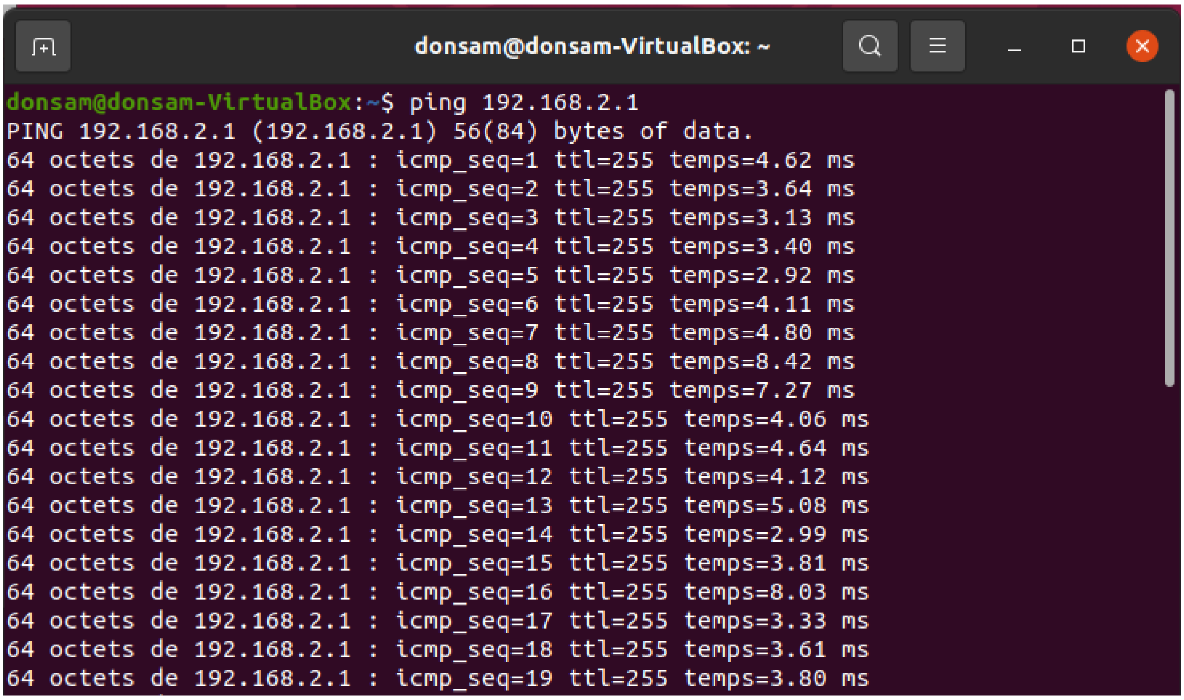
```
Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address.  Press <ENTER> for none:
> 192.168.2.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) y
```

```
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) y

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to LAN...
 Reloading filter...
 Reloading routing configuration...
 DHCPD...

The IPv4 LAN address has been set to 192.168.2.1/24


The IPv6 LAN address has been set to dhcp6

Press <ENTER> to continue.
```

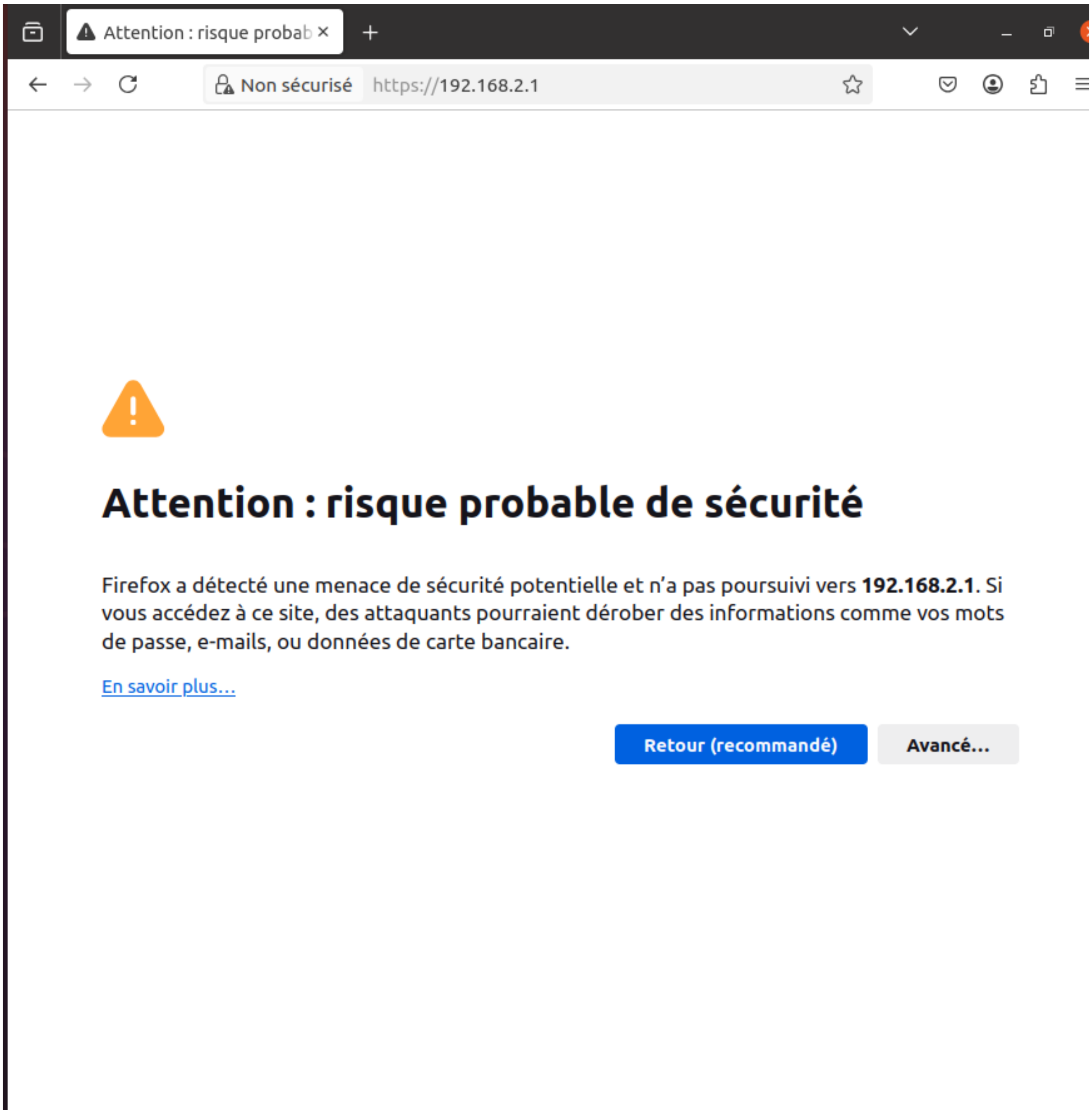
```
The IPv6 LAN address has been set to dhcp6

Press <ENTER> to continue.
VirtualBox Virtual Machine - Netgate Device ID: a11264d7bf28a86c2b31

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

 WAN (wan)       -> em0        -> v4/DHCP4: 10.0.2.15/24
                                  v6/DHCP6: fd17:625c:f037:2:a00:27ff:feed:afc
4
 LAN (lan)       -> em1        -> v4: 192.168.2.1/24

 0) Logout (SSH only)                  9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart webConfigurator
 3) Reset webConfigurator password    12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```

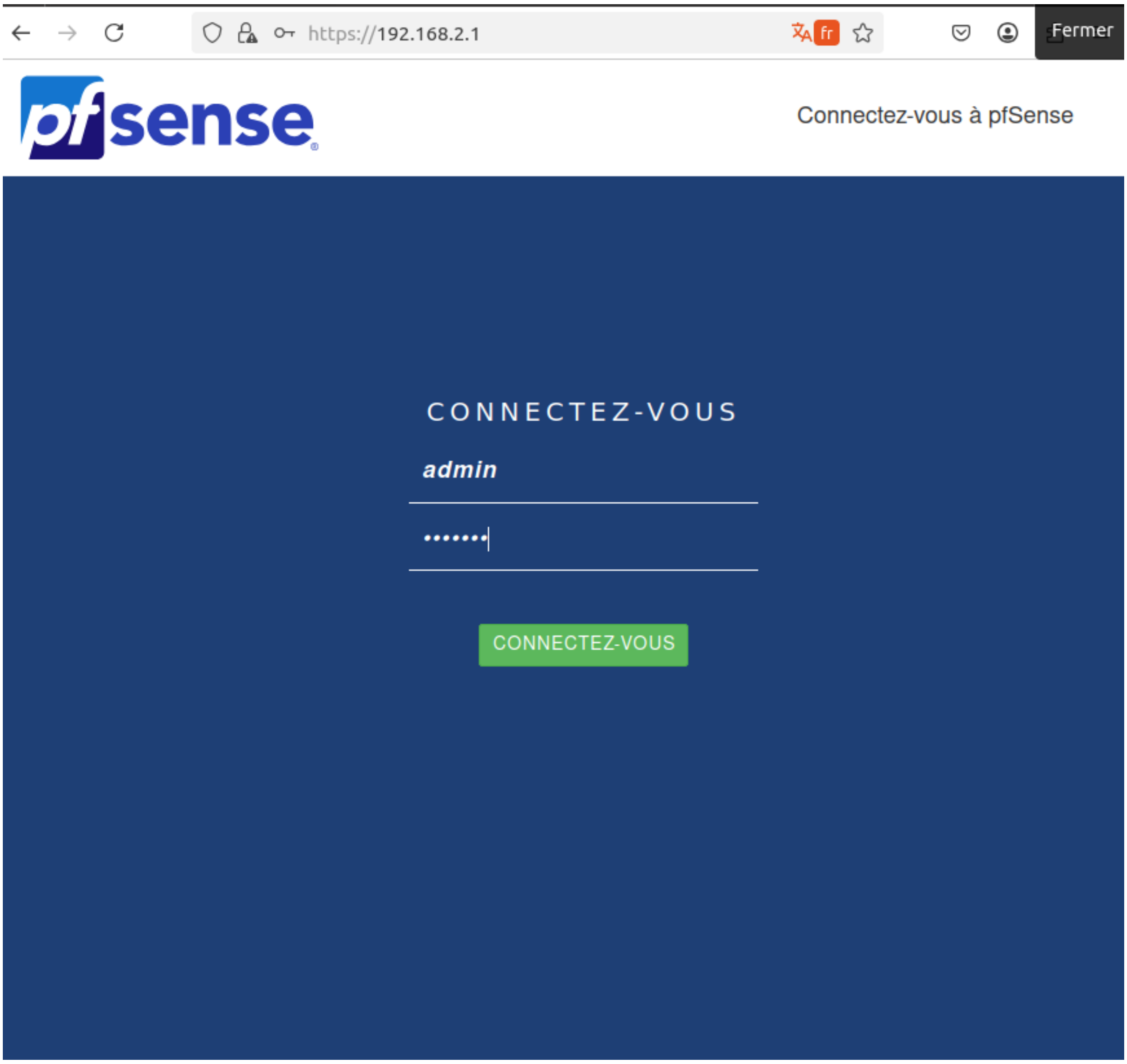
Sur Ubuntu-VM

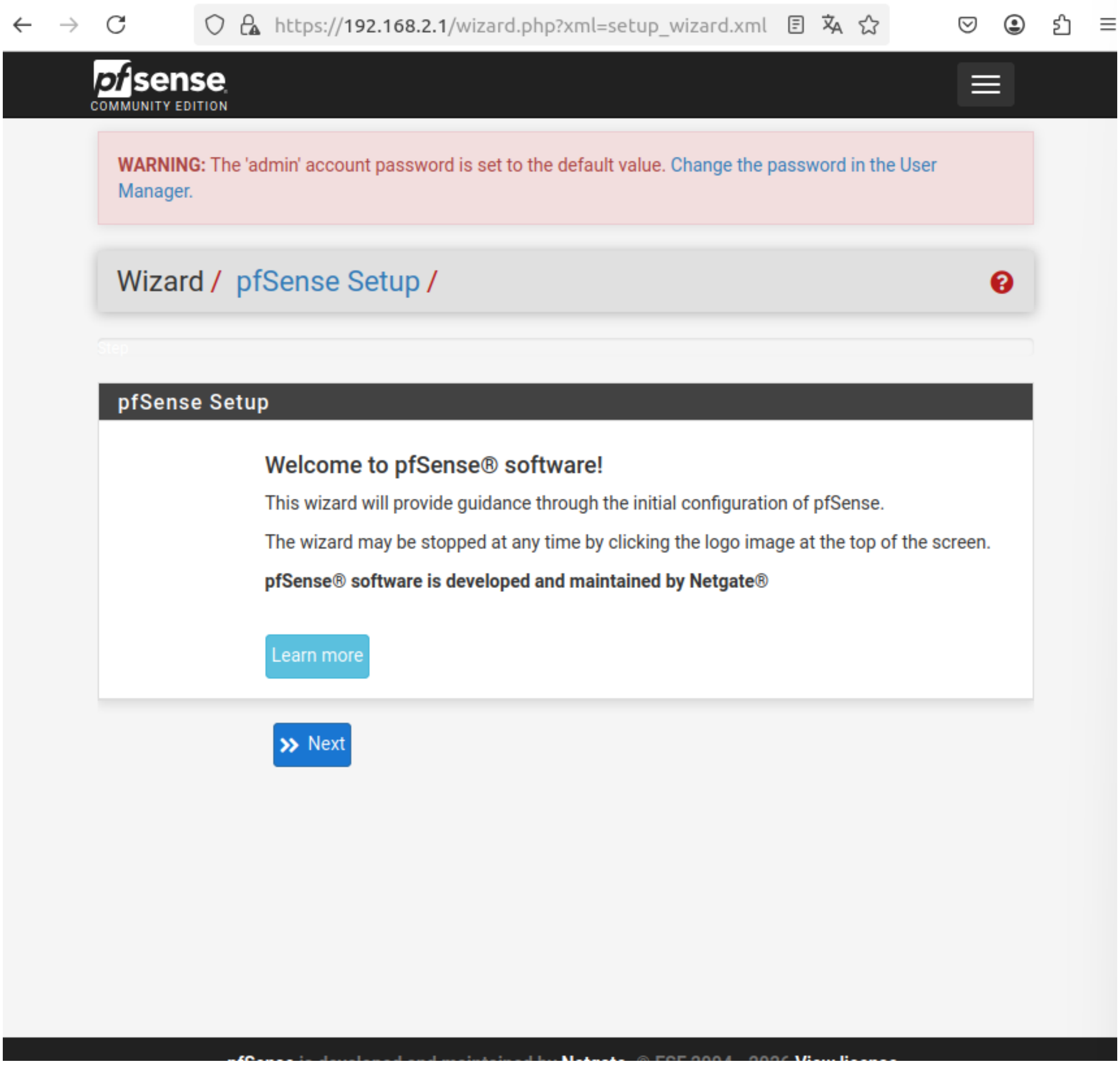




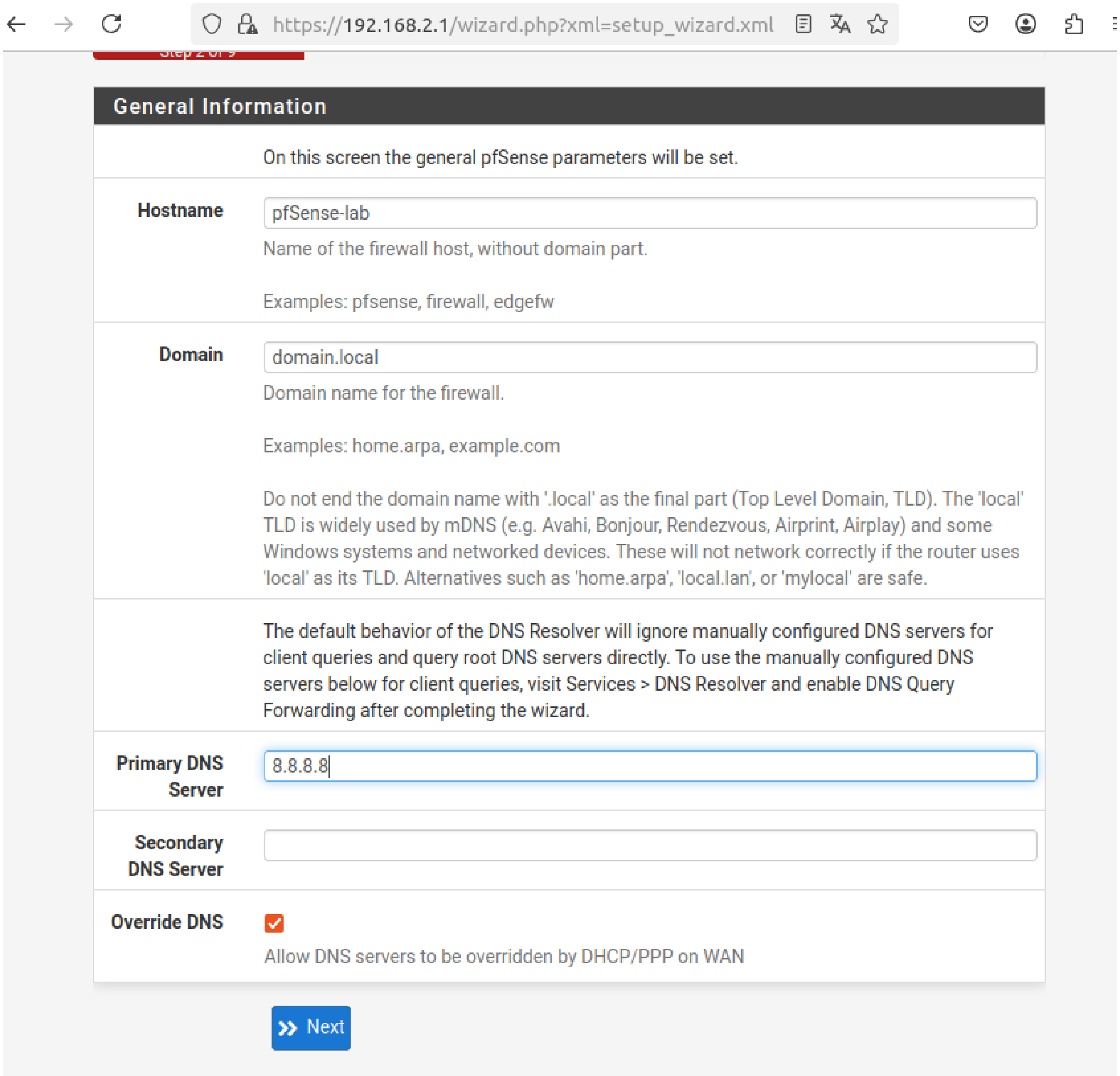
✓ **Accès à l'interface Web**

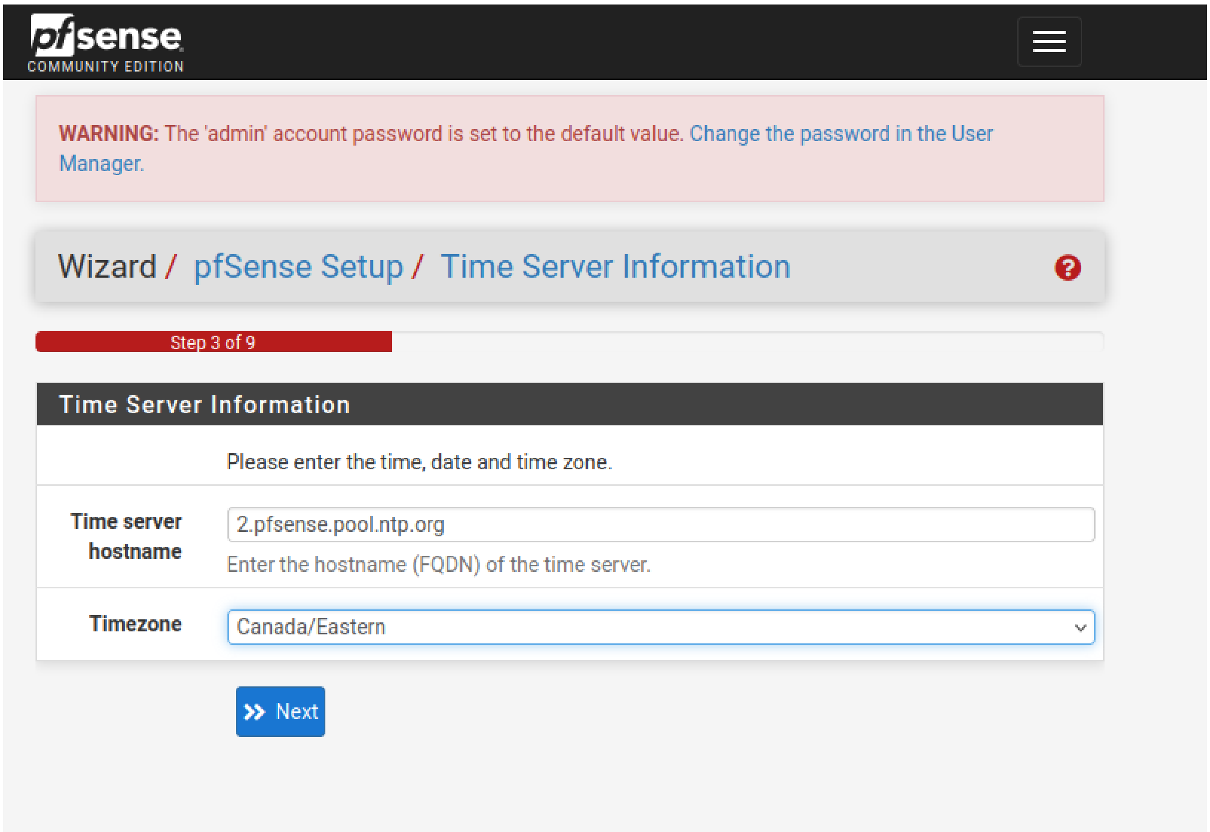


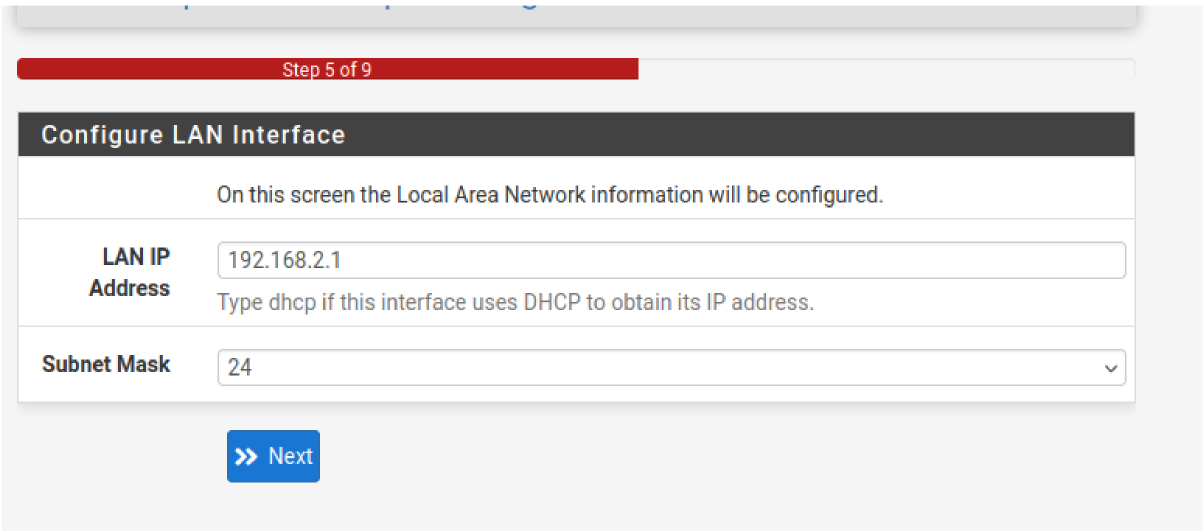

https://192.168.2.1
fr
Fermer
pfsense
Connectez-vous à pfSense
CONNECTEZ-VOUS
admin
•••••••
CONNECTEZ-VOUS

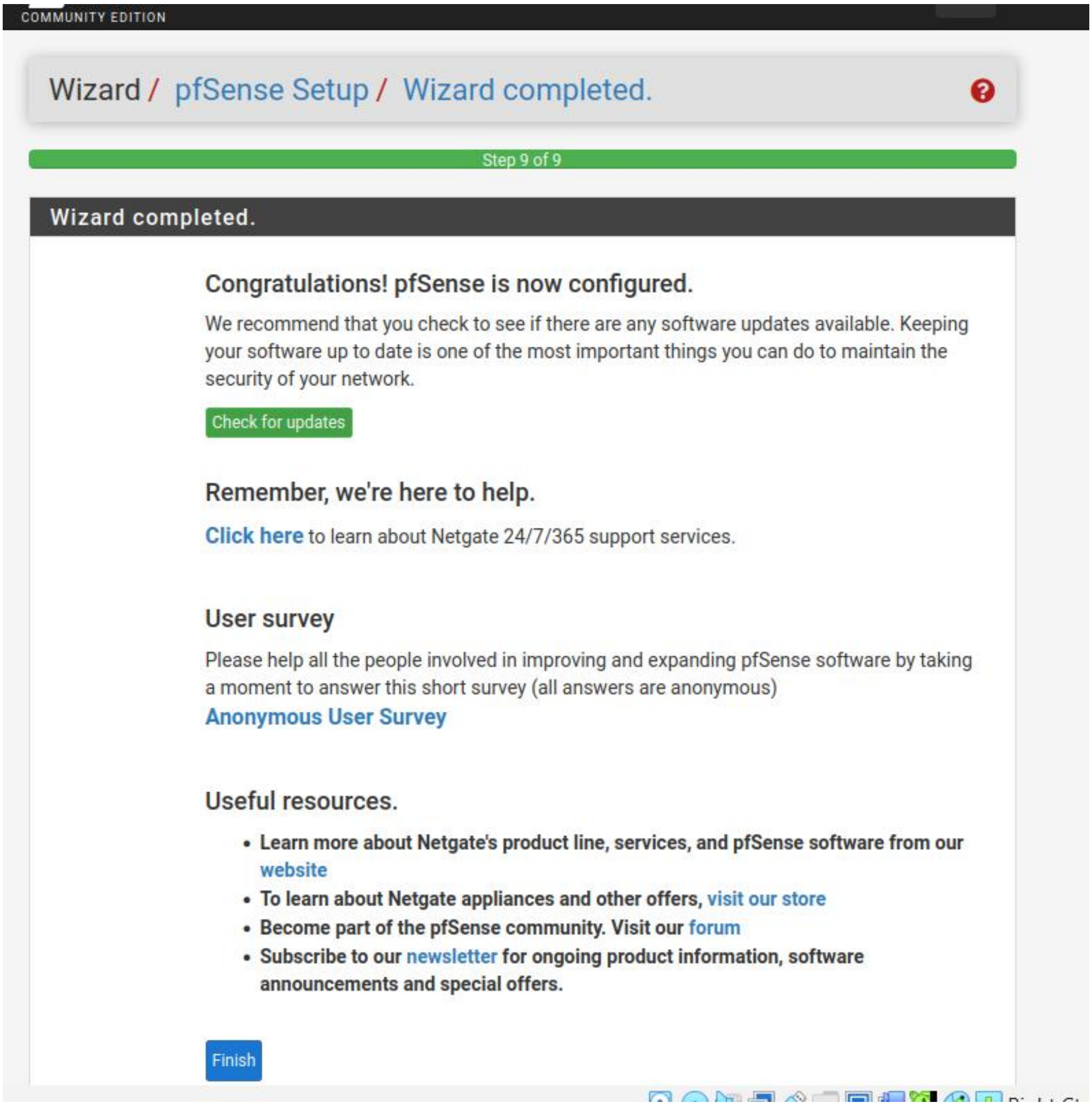

https://192.168.2.1/wizard.php?xml=setup_wizard.xml
pfsense
COMMUNITY EDITION
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Wizard / pfSense Setup /
pfSense Setup
Welcome to pfSense® software!
This wizard will provide guidance through the initial configuration of pfSense.
The wizard may be stopped at any time by clicking the logo image at the top of the screen.
pfSense® software is developed and maintained by Netgate®
Learn more
Next

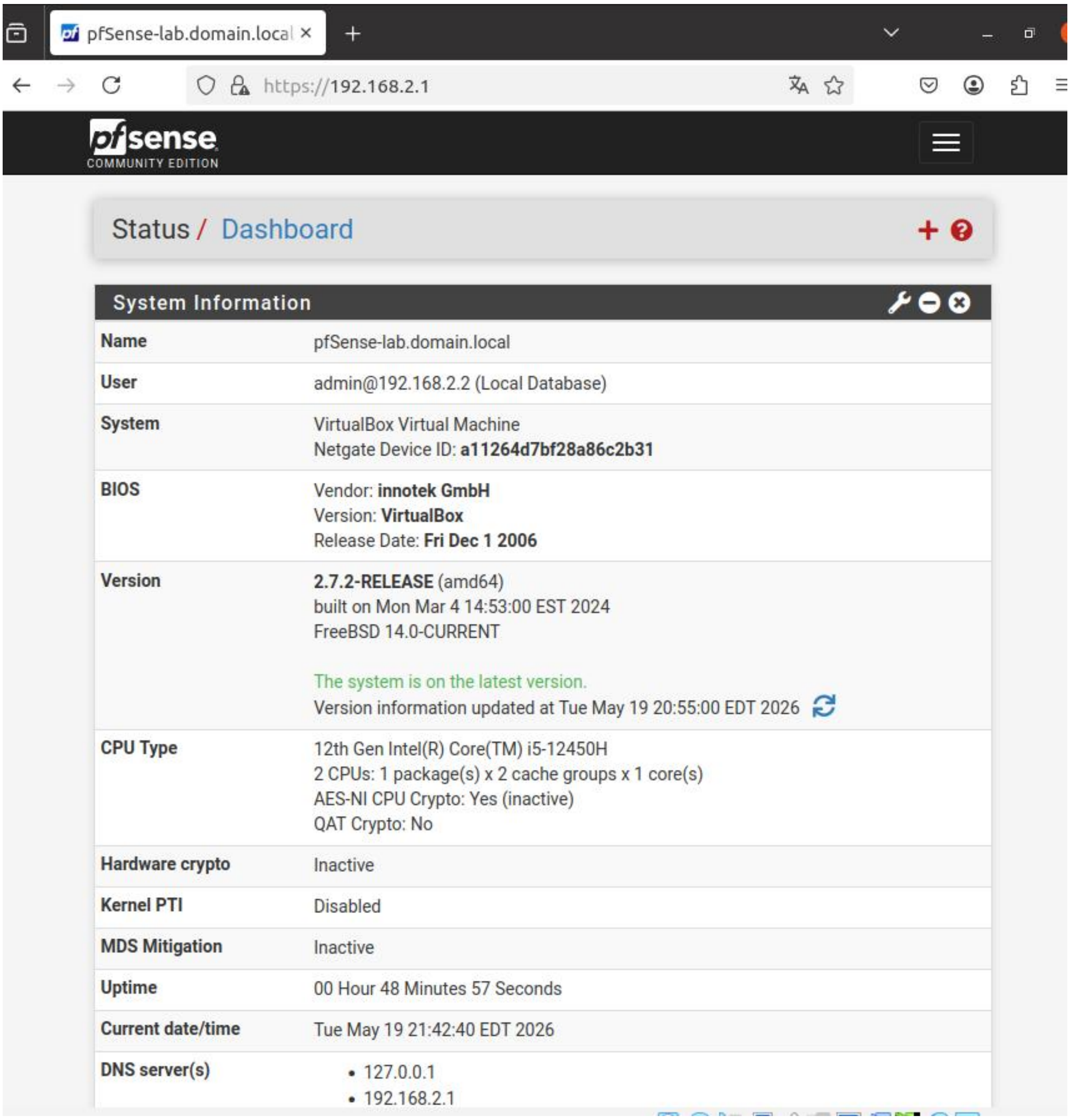
❖ CONFIGURATION DE BASE pfSense




pfsense
COMMUNITY EDITION
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Wizard / pfSense Setup / Time Server Information
Step 3 of 9
Time Server Information
Please enter the time, date and time zone.
Time server hostname
2.pfsense.pool.ntp.org
Enter the hostname (FQDN) of the time server.
Timezone
Canada/Eastern
Next

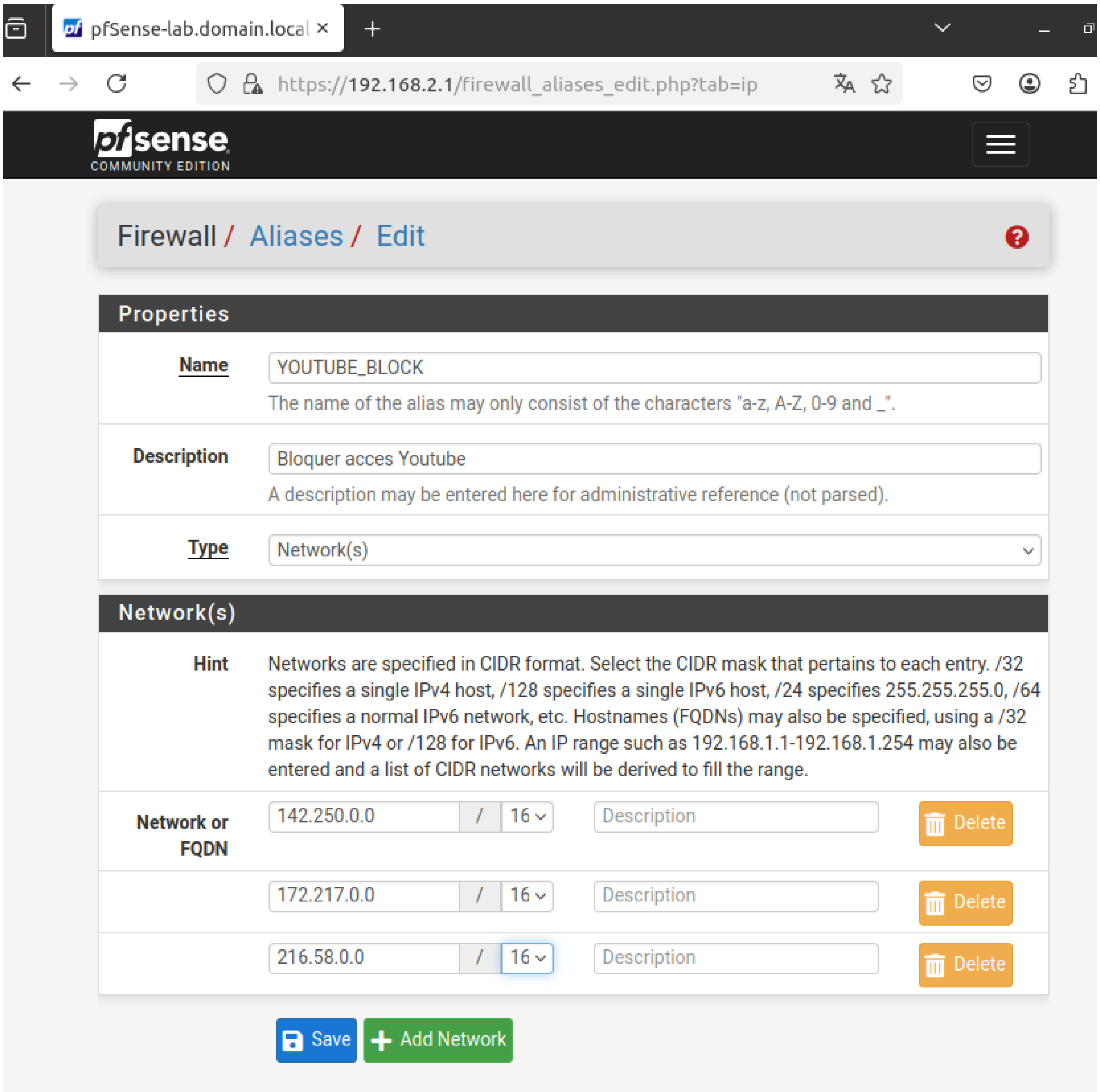

Step 5 of 9
Configure LAN Interface
On this screen the Local Area Network information will be configured.
LAN IP Address
192.168.2.1
Type dhcp if this interface uses DHCP to obtain its IP address.
Subnet Mask
24
Next


COMMUNITY EDITION
Wizard / pfSense Setup / Wizard completed.
Step 9 of 9
Wizard completed.
Congratulations! pfSense is now configured.
We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.
Check for updates
Remember, we're here to help.
Click here to learn about Netgate 24/7/365 support services.
User survey
Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)
Anonymous User Survey
Useful resources.
• Learn more about Netgate's product line, services, and pfSense software from our website
• To learn about Netgate appliances and other offers, visit our store
• Become part of the pfSense community. Visit our forum
• Subscribe to our newsletter for ongoing product information, software announcements and special offers.
Finish

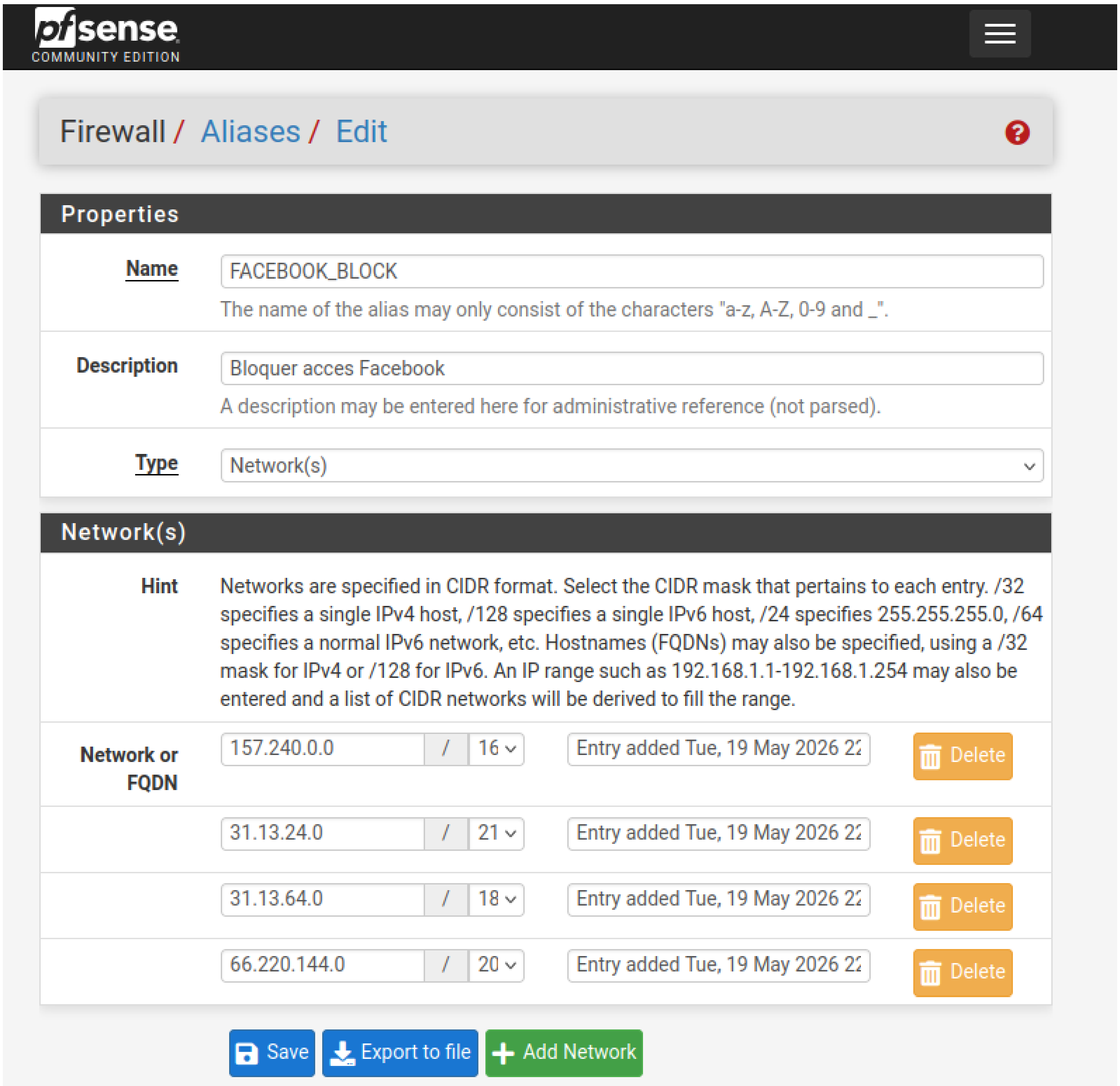

pfSense-lab.domain.local
https://192.168.2.1
pfsense
COMMUNITY EDITION
Status / Dashboard
System Information
Name
pfSense-lab.domain.local
User
admin@192.168.2.2 (Local Database)
System
VirtualBox Virtual Machine
Netgate Device ID: a11264d7bf28a86c2b31
BIOS
Vendor: innotek GmbH
Version: VirtualBox
Release Date: Fri Dec 1 2006
Version
2.7.2-RELEASE (amd64)
built on Mon Mar 4 14:53:00 EST 2024
FreeBSD 14.0-CURRENT
The system is on the latest version.
Version information updated at Tue May 19 20:55:00 EDT 2026
CPU Type
12th Gen Intel(R) Core(TM) i5-12450H
2 CPUs: 1 package(s) x 2 cache groups x 1 core(s)
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto
Inactive
Kernel PTI
Disabled
MDS Mitigation
Inactive
Uptime
00 Hour 48 Minutes 57 Seconds
Current date/time
Tue May 19 21:42:40 EDT 2026
DNS server(s)
127.0.0.1
192.168.2.1

CONFIGURATION FIREWALL

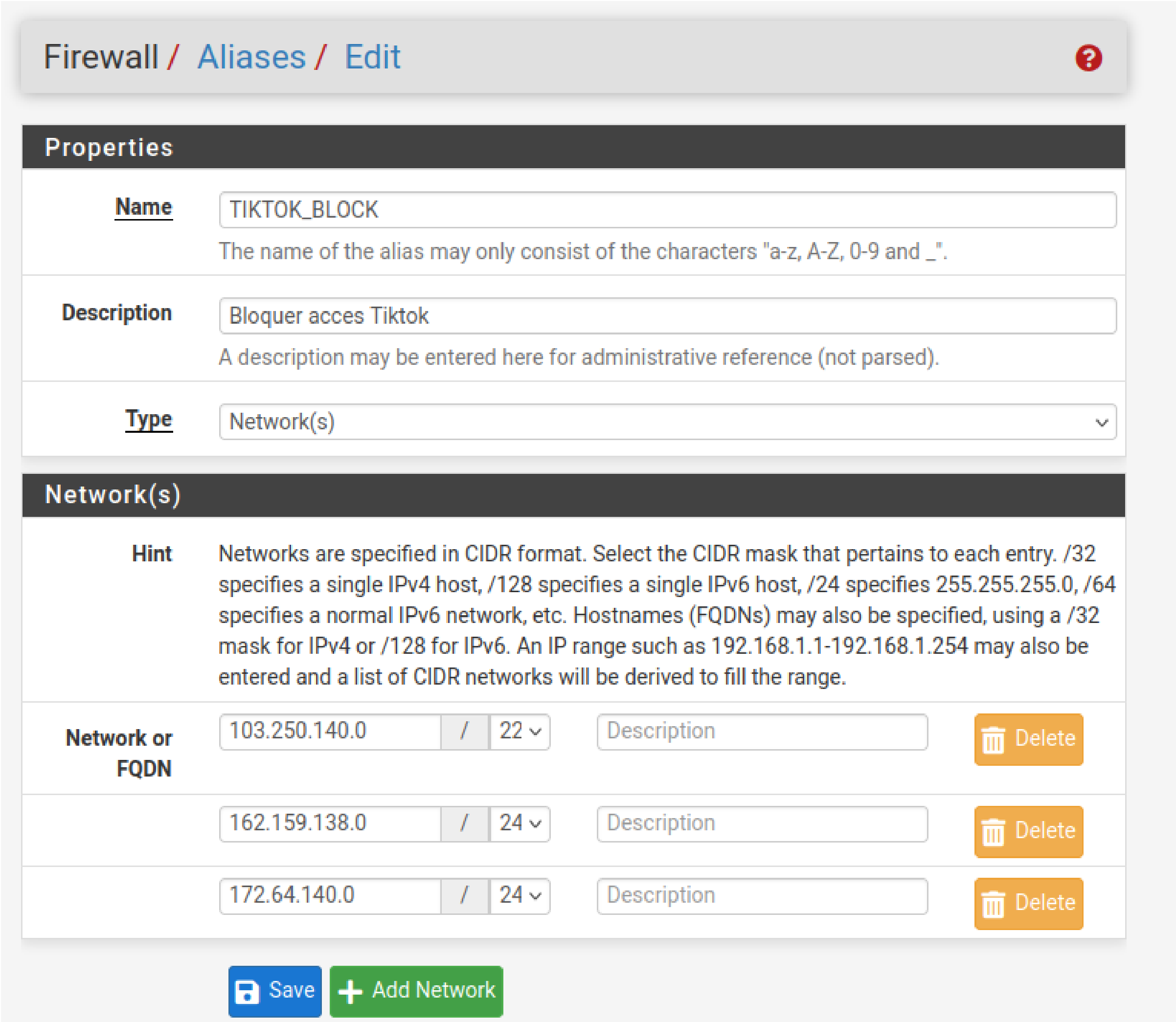
➢ **Bloquer l'accès à YouTube par IP sur pfSense**

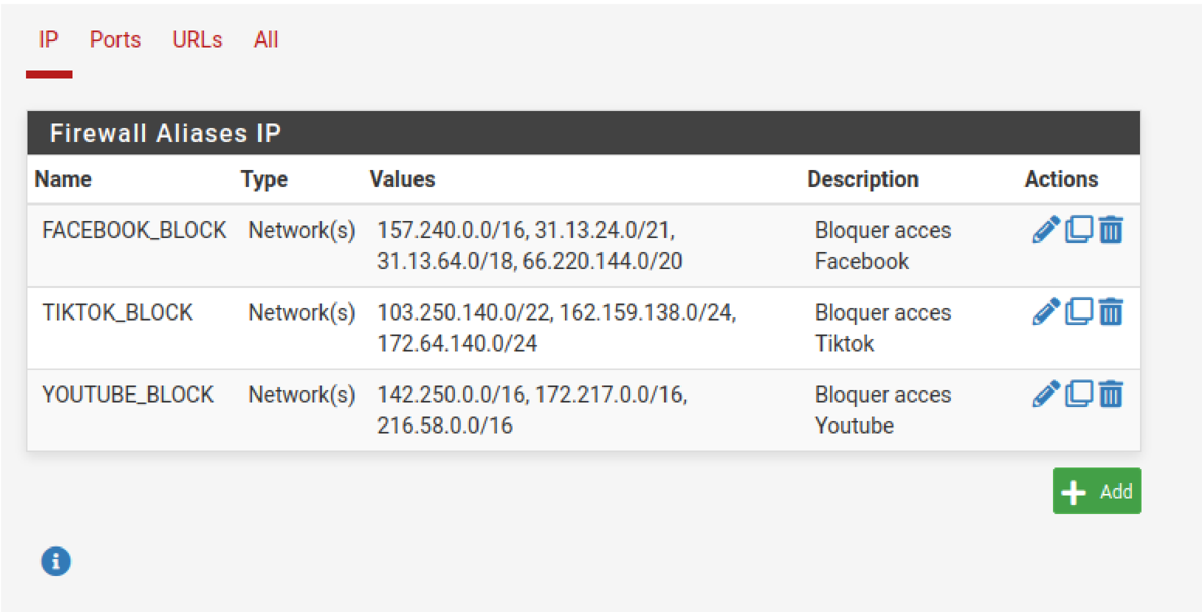


➢ **Bloquer l'accès à FaceBook par IP sur pfSense**

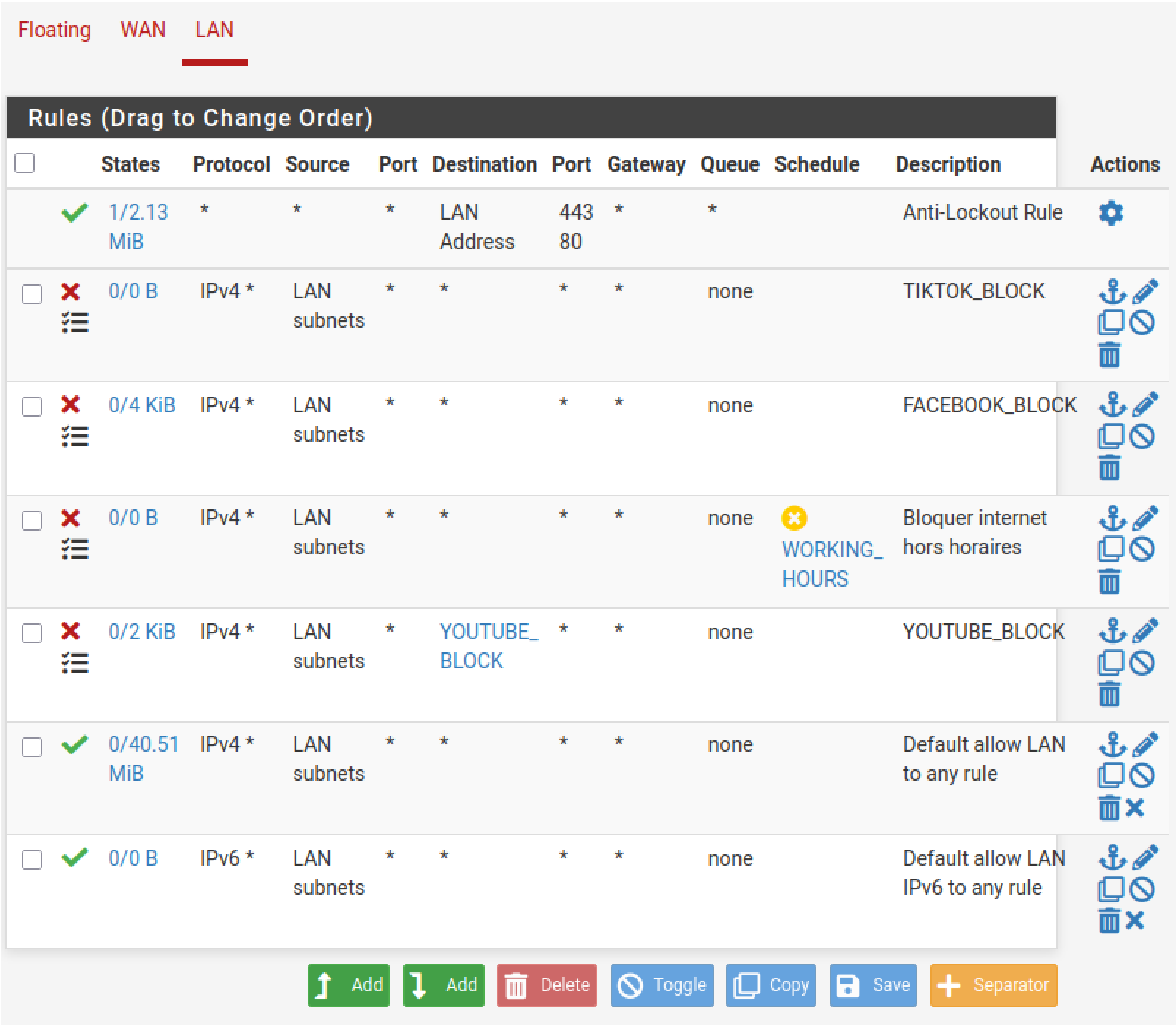


- **Bloquer l'accès à Tiktok par IP sur pfSense**





➢ **Règles Firewall pour YouTube, Facebook et TikTok**



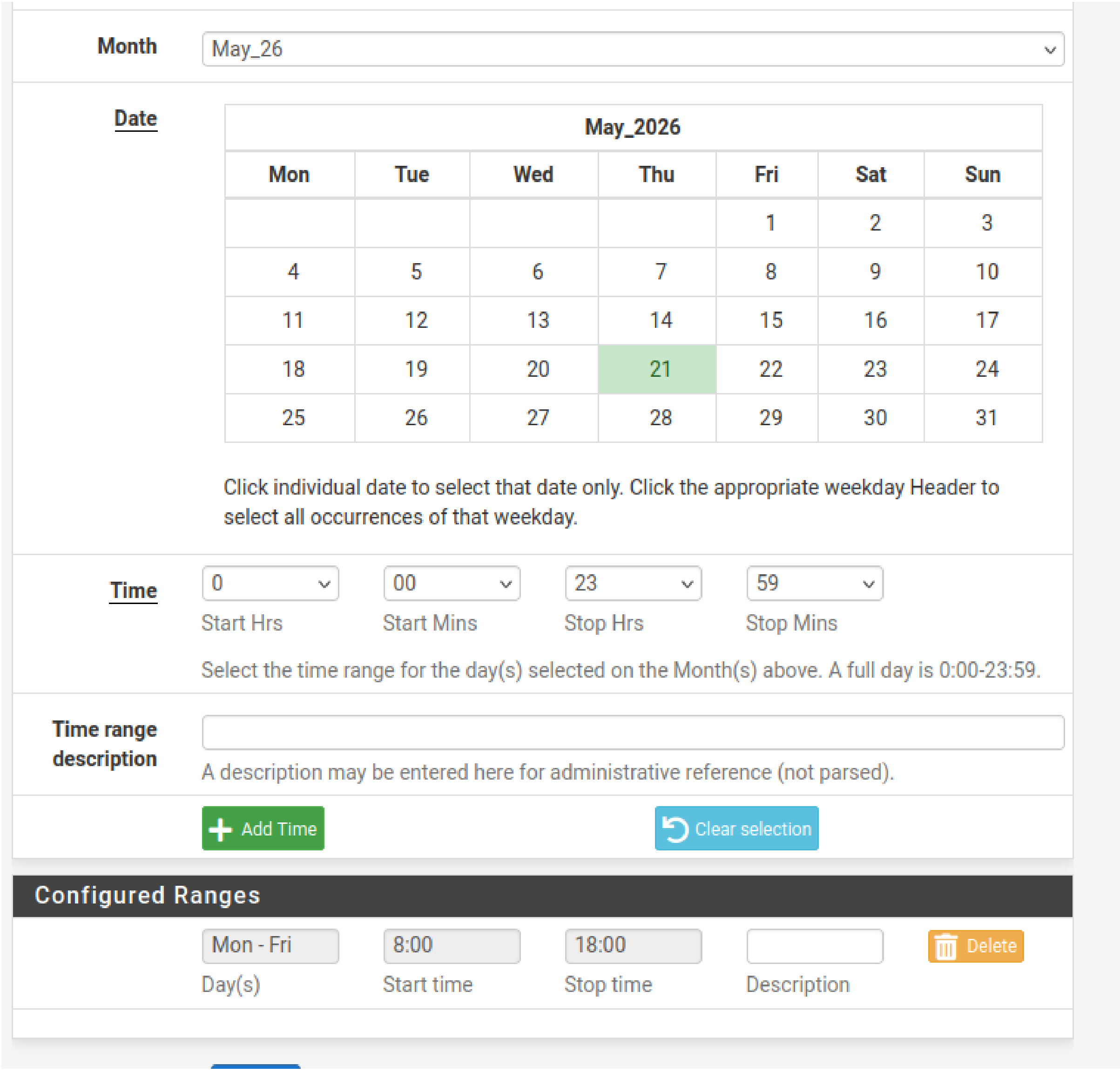
Floating WAN LAN

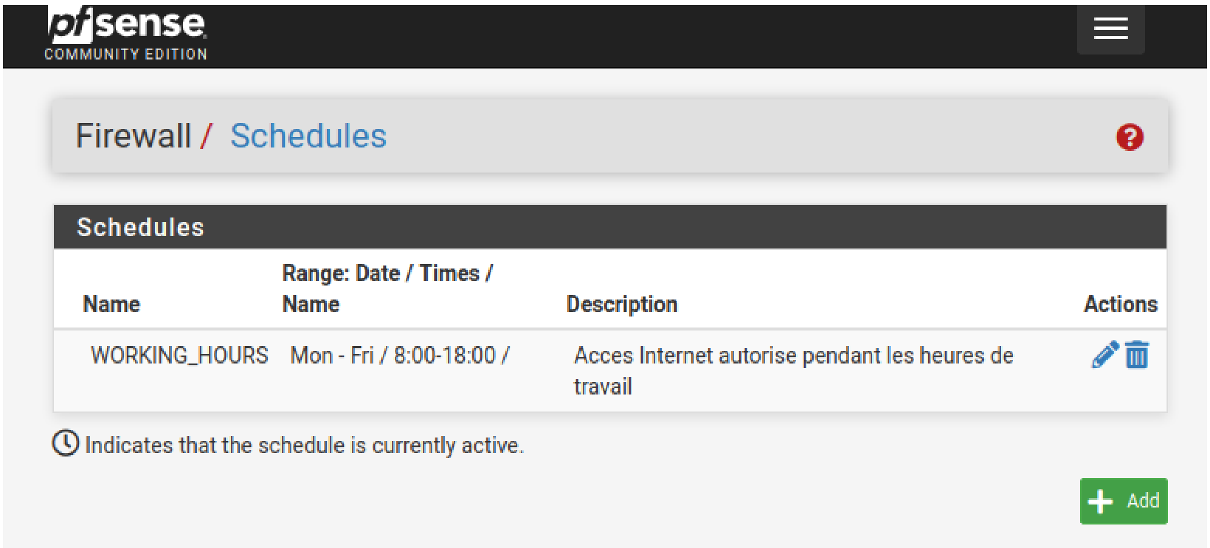
Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✔	1/2.13 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
✖	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		TIKTOK_BLOCK	
✖	0/4 KiB	IPv4 *	LAN subnets	*	*	*	*	none		FACEBOOK_BLOCK	
✖	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none	WORKING_ HOURS	Bloquer internet hors horaires	
✖	0/2 KiB	IPv4 *	LAN subnets	*	YOUTUBE_ BLOCK	*	*	none		YOUTUBE_BLOCK	
✔	0/40.51 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
✔	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

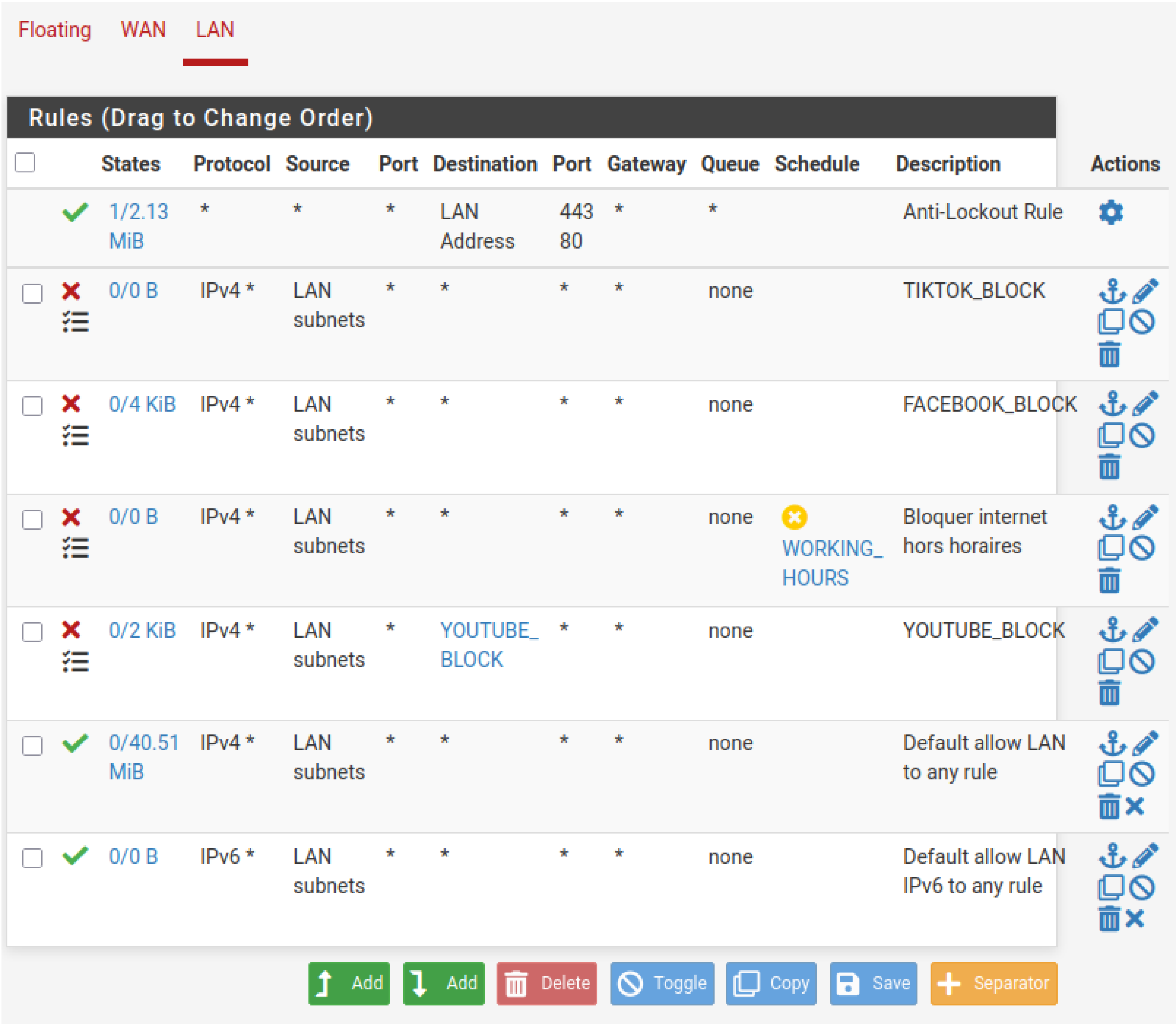
Add Add Delete Toggle Copy Save Separator

- **Création du Schedule (Planning horaire)**





➢ **Règle Firewall pour le Schedule**



Floating WAN LAN

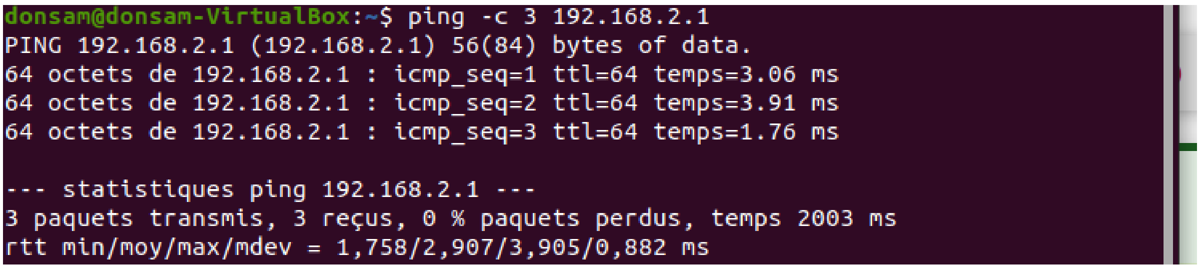
Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✔	1/2.13 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
✖	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		TIKTOK_BLOCK	
✖	0/4 KiB	IPv4 *	LAN subnets	*	*	*	*	none		FACEBOOK_BLOCK	
✖	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none	WORKING_HOURS	Bloquer internet hors horaires	
✖	0/2 KiB	IPv4 *	LAN subnets	*	YOUTUBE_BLOCK	*	*	none		YOUTUBE_BLOCK	
✔	0/40.51 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
✔	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add Add Delete Toggle Copy Save Separator

Test

✓ **TEST de Connectivité de base**

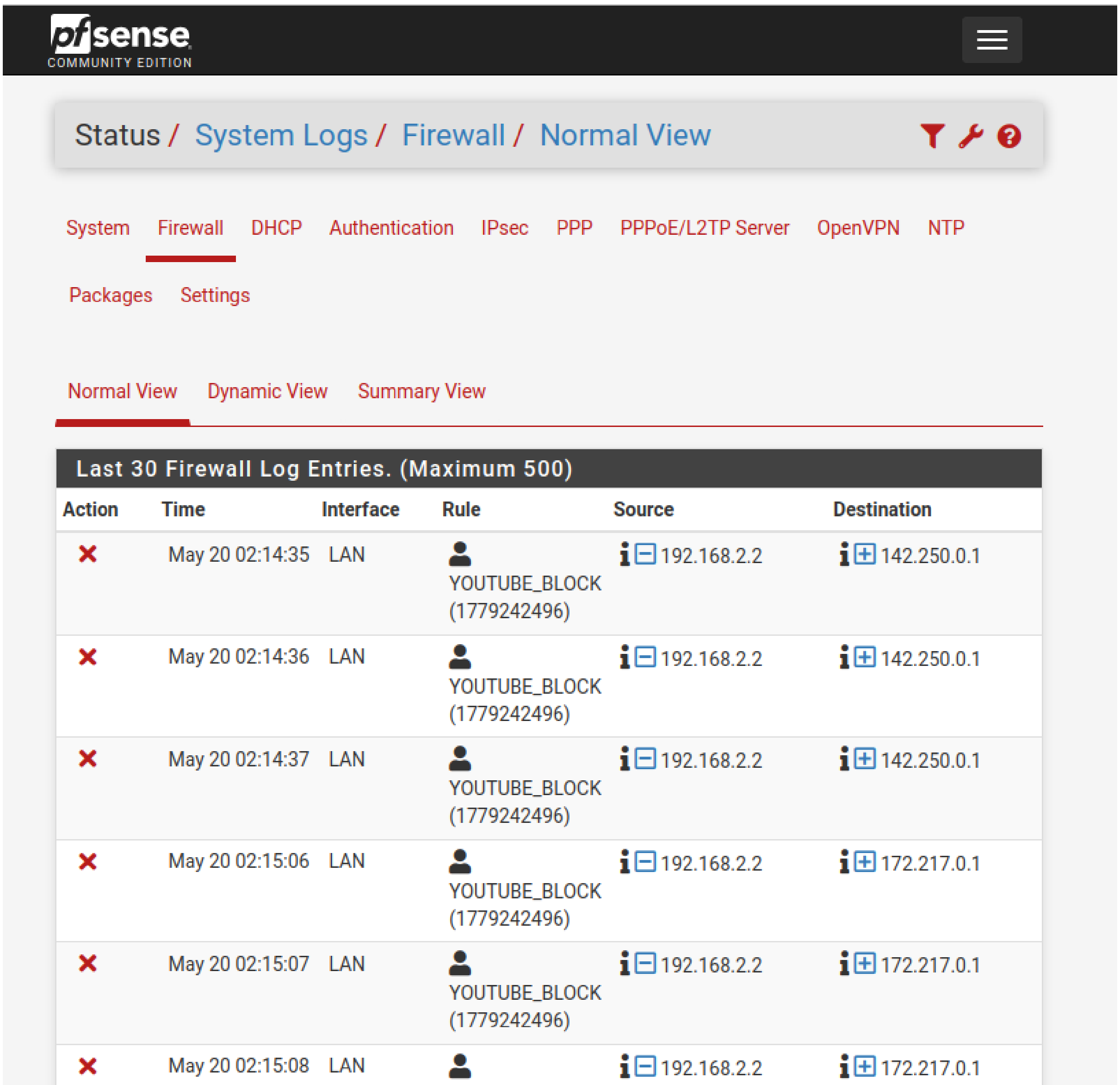


```
donsam@donsam-VirtualBox:~$ ping -c 3 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 octets de 192.168.2.1 : icmp_seq=1 ttl=64 temps=3.06 ms
64 octets de 192.168.2.1 : icmp_seq=2 ttl=64 temps=3.91 ms
64 octets de 192.168.2.1 : icmp_seq=3 ttl=64 temps=1.76 ms

--- statistiques ping 192.168.2.1 ---
3 paquets transmis, 3 reçus, 0 % paquets perdus, temps 2003 ms
rtt min/moy/max/mdev = 1,758/2,907/3,905/0,882 ms
```

- ✓ **TEST de Blocage YouTube par IP**

```
donsam@donsam-VirtualBox:~$ ping -c 3 142.250.0.1
PING 142.250.0.1 (142.250.0.1) 56(84) bytes of data.

--- statistiques ping 142.250.0.1 ---
3 paquets transmis, 0 reçus, 100 % paquets perdus, temps 2051 ms

donsam@donsam-VirtualBox:~$ ping -c 3 172.217.0.1
PING 172.217.0.1 (172.217.0.1) 56(84) bytes of data.

--- statistiques ping 172.217.0.1 ---
3 paquets transmis, 0 reçus, 100 % paquets perdus, temps 2036 ms

donsam@donsam-VirtualBox:~$ ping -c 3 216.58.0.1
PING 216.58.0.1 (216.58.0.1) 56(84) bytes of data.

--- statistiques ping 216.58.0.1 ---
3 paquets transmis, 0 reçus, 100 % paquets perdus, temps 2050 ms

donsam@donsam-VirtualBox:~$ 
```

- ✓ **TEST de vérification logs firewall**



Questions

1. **Quel est le rôle de pfSense ?**

R - pfSense est un pare-feu open source basé sur FreeBSD. Il sécurise le réseau interne, gère le trafic Internet, applique des règles de filtrage, fait du NAT et permet le contrôle des accès par horaires.

2. Différence WAN et LAN ?

- **WAN** (Wide Area Network) : interface connectée à Internet (côté externe)
- **LAN** (Local Area Network) : interface connectée au réseau interne (côté machines clientes)

2. **Pourquoi utiliser NAT automatique ?**

R - Le NAT (Network Address Translation) permet à toutes les machines du réseau interne (192.168.2.x) de partager une seule IP publique pour accéder à Internet. Sans NAT, les machines internes ne pourraient pas communiquer avec Internet.

3. **Comment fonctionne une règle firewall ?**

R - pfSense lit les règles de haut en bas et applique la première règle qui correspond au trafic. Chaque règle définit une Action (PASS/BLOCK), une Interface, un protocole, une source et une destination. Si aucune règle ne correspond, le trafic est bloqué par défaut.

4. **Pourquoi VirtualBox est utilisé pour les tests ?**

R - VirtualBox permet de créer un environnement réseau virtualisé complet sans matériel physique. On peut simuler plusieurs machines (pfSense, clients) interconnectées de manière isolée et sécurisée.

5. **Quel est le rôle de pfSense dans ce réseau ?**

R - Dans le réseau de l'IUS, pfSense est le pare-feu central qui contrôle tout le trafic entre le réseau du campus et Internet. Il filtre les accès, bloque les sites non pédagogiques, applique des horaires d'accès et journalise toutes les connexions.

6. **Pourquoi utiliser un Schedule ?**

R - Un Schedule permet d'automatiser l'activation et la désactivation des règles firewall selon des plages horaires définies. Cela évite une intervention manuelle et permet de gérer l'accès Internet selon les besoins (heures de cours, heures de bureau).

8. Différence entre PASS et BLOCK dans les logs ?

- **PASS** : le trafic est autorisé à passer par le firewall
- **BLOCK** : le trafic est bloqué et abandonné silencieusement sans notification à l'expéditeur

Conclusion

Ce TD nous a permis d'installer et configurer pfSense sur VirtualBox afin de sécuriser un réseau d'entreprise. Nous avons mis en place des règles de filtrage pour bloquer les sites non autorisés (YouTube, Facebook, TikTok) et un planning horaire pour contrôler l'accès à Internet. Cela démontre l'efficacité de pfSense comme solution de sécurité réseau open source accessible et facile à déployer.